7/5/2025 (E)

SOMAIYA VIDYAVIHAR UNIVERSITY

Semester: January 2025 –April 2025		
Maximum Marks: 100 Examination: ESE Examination		Duration:3 Hrs.
Programme code: 01 Programme: BTech Computer Engineering	Class: TY	Semester:VI(SVU 2020)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department: COMP	
Course Code: 11U01E626	Name of the Course: Machine Learning	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	**20**
i)	List and explain the key steps involved in selecting the appropriate Machine Learning algorithm for a given problem	5
ii)	Explain the working of the Random Forest algorithm. How does it reduce overfitting compared to a single decision tree?	5
iii)	What is model-based learning in the context of reinforcement learning? Explain with an example.	5
iv)	Explain Expectation-Maximization Algorithm	5
v)	Explain the Applications of reinforcement learning.	5
vi)	Explain multiclass classification with example	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	**10**
i)	Explain the working principle of Logistic Regression. How does it differ from Linear Regression in terms of prediction and loss function?	5
ii)	Explain Gaussian mixture model.	5
	OR	
Q2 A	Consider a dataset with two features X1 and X2 and 15 data points :	10

ID	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
X1	2	2	11	6	6	1	5	4	10	7	9	4	3	3	6
X2	10	6	11	9	4	2	10	9	12	5	11	6	10	8	11

Apply k-means algorithm to this dataset upto 2 iterations. Assume k=3 and data points with ID 2, 7 and 15 as the initial cluster centroids.

Q 2 B	Solve any **One**	**10**
i)	Company developing a Machine Learning model to predict whether a patient is at risk of developing diabetes based on a dataset containing information such as age, BMI, blood pressure, glucose levels, family history, and lifestyle habit. List and explain the data preprocessing steps you would perform on this dataset before applying any machine learning algorithm	10

ii)	What is the confusion matrix? Explain five performance measures of classification	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	**20**
i)	Following dataset of 6 students and their performance in a course. The goal is, to build a decision tree using the **CART algorithm** based on the Gini Index to classify whether a student passes or fails.	10

Student	Hours Studied	Attendance (%)	Pass/Fail
A	High	High	Pass
B	Low	High	Fail
C	High	Low	Pass
D	Low	Low	Fail
E	High	High	Pass
F	Low	High	Fail

a) Calculate the **Gini Index** for the root node. *(2 marks)*
b) Compute the Gini Index for splitting the data on the attribute **"Hours Studied"**. *(3 marks)*
c) Compute the Gini Index for splitting on **"Attendance"**. *(3 marks)*
d) Based on the Gini values, determine the best attribute to split on and justify your answer. *(2 marks)*

ii) Following dataset representing the relationship between **years of experience (X)** and **annual salary (Y)** for a group of employees: 10

X (Years of Experience)	Y (Annual Salary in $1000)
2	35
6	60
4	50
1	30
5	55
3	40

a) Using the **least squares method**, calculate the **slope (m)** and **intercept (b)** of the linear regression line. Show all steps clearly. *(5 marks)*

b) Calculate the **coefficient of determination (R^2)** for this linear regression model. What does this value indicate about the model's performance? *(3 marks)*

c) Based on the regression equation, **predict the salary** for an employee with **7 years of experience**. *(2 marks)*

iii)	What is the role of an activation function in a neural network? Describe the different types of activation functions commonly used, explaining each in detail	10

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	**20**
i)	Explain the architecture of a multilayer neural network and describe the backpropagation algorithm used for training such networks in detail	10
ii)	Explain the Backpropagation algorithm used for training it in detail and Describe the structure and working of a Feed-Forward Neural Network	10
iii)	Describe how Principal Component Analysis (PCA) is used for dimensionality reduction. Explain its working principle in detail	10

Que. No.	Question	Max. Marks
Q5	Solve any **four**	**20**
i)	Differentiate Supervised and Unsupervised ML algorithm	5
ii)	Data Splitting for training and evaluation sets	5
iii)	Distance measures used in Clustering	5
iv)	Temporal Difference Learning	5
v)	Applications of Bayesian Belief networks	5
vi)	Radial Basis functions	5